

Ex. No: 9 Threading

PROGRAM: THREAD CREATION USING PTHREAD

C PROGRAM

```
#include <stdio.h>

#include <stdlib.h>

#include <pthread.h>

#include <unistd.h>

void *thread_function(void *arg)

{

    int i;

    for(i = 1; i <= 5; i++)

    {

        printf("Thread Executing : %d\n", i);

        sleep(1);

    }

    pthread_exit(NULL);

}

int main()

{

    pthread_t t1, t2;

    pthread_create(&t1, NULL, thread_function, NULL);

    pthread_create(&t2, NULL, thread_function, NULL);

    pthread_join(t1, NULL);

    pthread_join(t2, NULL);

    printf("All Threads Completed\n");

    return 0;

}
```

Output:

```
(kali㉿kali)-[~]
└─$ ./thread
Thread 2 : 1
Thread 1 : 1
Thread 2 : 2
Thread 1 : 2
Thread 2 : 3
Thread 1 : 3
Thread 2 : 4
Thread 1 : 4
Thread 1 : 5
Thread 2 : 5
All Threads Completed
```

Enter thread/process numbers (if any):

0: Bash

1: ./thread

Press Enter.

Search command:

Ask anything

SHELL SCRIPT

task1()

```
{
for i in 1 2 3 4 5
do
echo "Thread 1 : $i"
sleep 1
done
}
```

task2()

```
{
for i in 1 2 3 4 5
do
echo "Thread 2 : $i"
sleep 1
done
}
```

task1 &

task2 &

wait

echo "All Threads Completed"

Output:

```
(kali㉿kali)-[~] -- number of processes:
$ chmod +x thread.sh
```

```
(kali㉿kali)-[~] -- back processes:
$ ./thread.sh
Thread 1 : 1
Thread 2 : 1
Thread 2 : 2
Thread 1 : 2
Thread 1 : 3
Thread 2 : 3
Thread 1 : 4
Thread 2 : 4
Thread 1 : 5
Thread 2 : 5
All Threads Completed
```

```
Thread 1 : 1
Thread 2 : 1
Thread 1 : 2
Thread 2 : 2
Thread 1 : 3
Thread 2 : 3
Thread 1 : 4
Thread 2 : 4
Thread 1 : 5
Thread 2 : 5
Ask anything
```